Python Lab Assignment Solutions

1. Plot a graph of $y = x^2$ for x values from 0 to 10 with customization.

```
import matplotlib.pyplot as plt
import numpy as np

x = np.arange(0, 11)  # x values from 0 to 10
y = x**2  # y = x^2

plt.plot(x, y, color='red', linestyle='--')
plt.title("Quadratic Function")
plt.xlabel("x")
plt.ylabel("y")
plt.show()
```

2. Plot a bar chart of students' scores.

```
import matplotlib.pyplot as plt

students = ["Alice", "Bob", "Charlie", "David"]
scores = [85, 92, 78, 90]

plt.bar(students, scores, color='blue')
plt.title("Student Scores")
plt.xlabel("Students")
plt.ylabel("Scores")
plt.show()
```

3. Generate random x and y data points and create a scatter plot.

```
import numpy as np
import matplotlib.pyplot as plt

x = np.random.randint(1, 50, 10)  # 10 random x points
y = np.random.randint(1, 50, 10)  # 10 random y points

plt.scatter(x, y, color='green')
plt.title("Random Scatter Plot")
plt.xlabel("X values")
plt.ylabel("Y values")
plt.show()
```

4. Multiply two 3x3 matrices using numpy.dot().

```
import numpy as np

A = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
B = np.array([[9, 8, 7], [6, 5, 4], [3, 2, 1]])

result = np.dot(A, B)
print("Result of Matrix Multiplication:\n", result)
```

5. Create and slice a NumPy array.

```
import numpy as np

arr = np.arange(1, 21)  # Array from 1 to 20

print("First 5 elements:", arr[:5])
print("Last 5 elements:", arr[-5:])
print("Elements at even indices:", arr[::2])
```

6. Perform operations on two NumPy arrays.

```
import numpy as np

a = np.array([1, 2, 3])
b = np.array([4, 5, 6])

print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
```

7. Calculate statistics of a NumPy array.

```
import numpy as np

data = np.array([10, 15, 7, 22, 17])

print("Mean:", np.mean(data))
print("Standard Deviation:", np.std(data))
print("Median:", np.median(data))
print("Sum of all elements:", np.sum(data))
```

8. Create and manipulate a DataFrame.

```
import pandas as pd

data = {
```

```python
    "Name": ["John", "Anna", "Peter", "Linda"],
    "Age": [28, 24, 35, 32],
    "City": ["New York", "London", "Berlin", "Tokyo"]
}

df = pd.DataFrame(data)

# Print the "Age" column
print("Age Column:\n", df["Age"])

# Add a new column "Salary"
df["Salary"] = [50000, 60000, 55000, 65000]

# Filter rows where "Age" > 30
filtered_df = df[df["Age"] > 30]
print("Filtered Rows (Age > 30):\n", filtered_df)
```

9. Merge two DataFrames.

```python
import pandas as pd

# DataFrame 1: Employee details
df1 = pd.DataFrame({
    "ID": [1, 2, 3, 4],
    "Name": ["John", "Anna", "Peter", "Linda"],
    "Department": ["HR", "IT", "Finance", "Marketing"]
})

# DataFrame 2: Employee salaries
df2 = pd.DataFrame({
    "ID": [1, 2, 3, 4],
    "Salary": [50000, 60000, 55000, 65000]
})

# Merge on "ID" column
merged_df = pd.merge(df1, df2, on="ID")
print("Merged DataFrame:\n", merged_df)
```

10. Read a CSV file, calculate average temperature, and plot a line graph.

```python
import pandas as pd
import matplotlib.pyplot as plt

# Assuming the CSV file has columns: 'Date', 'Temperature'
df = pd.read_csv('weather_data.csv')
```

```
# Convert "Date" column to datetime
df['Date'] = pd.to_datetime(df['Date'])

# Group by month and calculate average temperature
df['Month'] = df['Date'].dt.to_period('M')
monthly_avg = df.groupby('Month')['Temperature'].mean()

# Plot the line graph
monthly_avg.plot(kind='line', marker='o', color='blue')
plt.title("Average Monthly Temperature")
plt.xlabel("Month")
plt.ylabel("Average Temperature")
plt.show()
```